Solutions

Math 372: Midterm Exam #2

Wednesday, April 1

Instructions: You have 50 minutes. Calculators and notes are not allowed. You must show your work to recieve credit.

Problem 1. A service station has both self-service and full-service islands. On each island, there is a single regular unleaded pump with two hoses. Let X denote the number of hoses being used on the self-service island at a particular time, and let Y denote the number of hoses on the full-service island in use at that time. The joint pmf of X and Y is described by the following table:

$p(x,y)$	$y=0$	$y=1$	$y=2$	
$x=0$	0.10	0.08	0.06	.24
$x=1$	0.04	0.20	0.14	.38
$x=2$	0.02	0.06	0.30	.38
	.16	.34	.50	

2 pts (a) Find $\mathbb{P}[X = 1 \text{ and } Y = 1]$.

$\boxed{.20}$

2 pts (b) Find $\mathbb{P}[X \leq 1 \text{ and } Y \leq 1]$.

$\boxed{.42}$

2pts (c) Using words, describe the event $[X \neq 0 \text{ and } Y \neq 0]$, and then compute its probability.

At least one hose is in use at both statlers.

$.2 + .14 + .06 + .30 = \boxed{.70}$

2pts (d) Compute the marginal pmf of X and use it to calculate $\mathbb{P}[X \leq 1]$.

x	$p(x)$
0	.24
1	.38
2	.38

$P[X \leq 1] = .24 + .38 = \boxed{.62}$

2pts (e) Are X and Y independent random variables? Justify your answer.

y	$p(y)$
0	.16
1	.34
2	.50

$\boxed{\text{No}}$ $p(0,0) = .10 \neq (.16)(.24) = p_X(0)p_Y(0)$

Problem 2. Suppose X is a random variable with probability density function

$$f(x) = \begin{cases} Cx^2 & : \quad 0 < x < 1 \\ 0 & : \quad \text{otherwise} \end{cases}$$

(a) Find a value of C so that f is a valid pdf.

(b) Find the cumulative density function $F(t)$. *(Hint: don't forget to specify the value of $F(t)$ when $t \leq 0$ and when $t \geq 1$.)*

(c) What is $\mathbb{E}\left[(X+1)^2\right]$?

+3

(a) Need $\int_0^1 Cx^2\,dx = 1 \iff C\left[\frac{x^3}{3}\right]_0^1 = 1 \iff \frac{C}{3} = 1 \iff \boxed{C=3}$

(b) If $t \in (0,1)$, $F(t) = \int_0^t 3x^2\,dx = x^3\Big]_0^t = t^3$

If $t \leq 0$ $F(t) = 0$

If $t \geq 1$ $F(t) = 1$.

So $F(t) = \begin{cases} 0 : t \leq 0 & +1 \\ t^3 : 0 < t < 1 & +2 \\ 1 : t \geq 1 & +1 \end{cases}$

(c) $E[(X+1)^2] = \int_0^1 (x+1)^2\, 3x^2\,dx$ +2

$= 3\int_0^1 (x^2+2x+1)x^2\,dx$

$= 3\int_0^1 (x^4 + 2x^3 + x^2)\,dx$

$= 3\left[\frac{x^5}{5} + \frac{2}{4}x^4 + \frac{1}{3}x^3\right]_0^1$

$= 3\left(\frac{1}{5} + \frac{1}{2} + \frac{1}{3}\right)$ +1

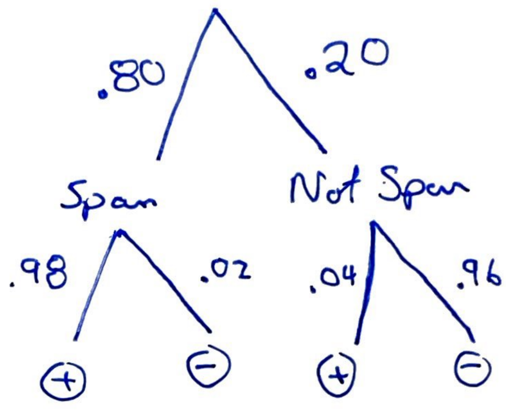
$= \frac{31}{10}$

$= \boxed{3.1}$

$\frac{7}{10} + \frac{1}{3} = \frac{21+10}{30} = \frac{31}{30}$

Problem 3. An email provider observes that 80% of emails are spam. It intends to filter spam emails before they reach the inbox. Spams emails are correctly tagged as spam 98% of the time, and the chance of tagging a non-spam email as spam is 4%.

(a) Draw a tree diagram to illustrate this situation.



(b) If a certain email is tagged as spam, find the probability that it is not a spam email. That is, find

$$\mathbb{P}\left[\text{email is not spam} \mid \text{email is tagged as spam}\right].$$

You don't need to simplify your answer.

$$P[\text{Not Spam} \mid +] = \frac{P[\text{Not Spam and } +]}{P[+]}$$

$$= \frac{(.2)(.04)}{(.2)(.04)+(.8)(.98)}$$

Problem 4. A measurement error X for a heat sensor has cdf

$$F(x) = \begin{cases} 0 & : \quad x < -2 \\ \frac{1}{2} + \frac{3}{32}\left(4x - \frac{x^3}{3}\right) & : \quad -2 \le x < 2 \\ 1 & : \quad x \ge 2 \end{cases}$$

1pts (a) Compute $\mathbb{P}[X < 0]$ $= F(0) = \frac{1}{2}$

2pts (b) Compute $\mathbb{P}[-1 < X < 1]$ $= F(1) - F(-1)$ +1

$$= \left[\frac{1}{2} + \frac{3}{32}\left(4 - \frac{1}{3}\right)\right] - \left[\frac{1}{2} + \frac{3}{32}\left(-4 + \frac{1}{3}\right)\right]$$

$$= \frac{3}{32}\left(4 - \frac{1}{3}\right) + \frac{3}{32}\left(4 - \frac{1}{3}\right) = \frac{3}{32} \cdot \frac{22}{3} = \frac{22}{32} = \frac{11}{16}$$ +1

4pts (c) Find the pdf of X. *[Hint: consider the three cases $x < -2$, $-2 \le x < 2$, and $x \ge 2$ separately].*

If $x \in [-2, 2]$, $f(x) = F'(x) = \frac{3}{32}(4 - x^2)$

So $F'(x) = \begin{cases} 0 & : \quad x < -2 & +1 \\ \frac{3}{32}(4 - x^2) & : \quad -2 \le x < 2 & +2 \\ 0 & : \quad x \ge 2 & +1 \end{cases}$

This is pdf $f(x)$

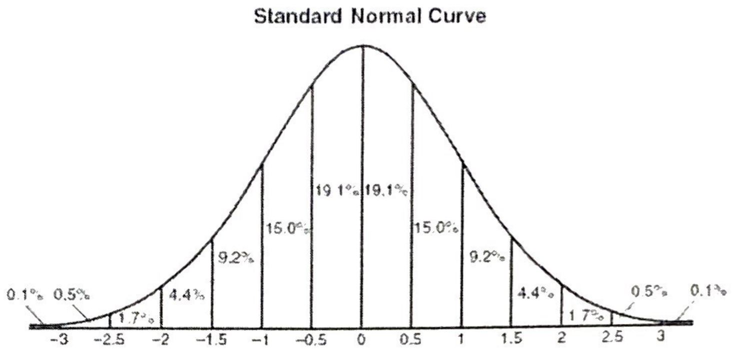
3pts (d) Verify that $\mathbb{E}[X] = 0$.

$$E[X] = \int_{-2}^{2} x f(x)\, dx = \frac{3}{32}\int_{-2}^{2}(4x - x^3)\, dx \quad +2$$

$$= \frac{3}{32}\left[2x^2 - \frac{x^4}{4}\right]_{-2}^{2} \quad +1$$

$$= 0$$

The density curve of the standard normal distribution is:



Problem 5. Resting breathing rates for college-age students are approximately normally distributed with mean 12 and standard deviation 2.3 breaths per minute. What fraction of all college-age students have breathing rates in the following intervals?

+5 (a) 9.7 to 14.3 breaths per minute

$$P[9.7 \leq X \leq 14.3] = P[-2.3 \leq X-12 \leq 2.3]$$
$$= P[-1 \leq Z \leq 1]$$
$$\approx 68.2\%$$

+5 (b) 9.7 to 16.6 breaths per minute

$$P[9.7 \leq X \leq 16.6] = P[-2.3 \leq X-12 \leq 4.6]$$
$$= P[-1 \leq Z \leq 2]$$
$$\approx 81.8\%$$

$$\begin{array}{r} 9.2 \\ 4.4 \\ + \; 68.2 \\ \hline 81.8 \end{array}$$